Tree Walk

Binary Trees, Inorder Traversal, Aggregation · Mentor-Led Lab (Student Edition)

Developed by TalenciaGlobal

Difficulty ★★ Intermediate

Problem

Build a binary tree from a level-order description, then walk it in order and summarise it.

Implement the build and the in-order traversal from scratch, reporting the values in order, the node count and their sum.

What to Compute

- Inorder: the values in in-order (left, node, right).
- Nodes: the total number of nodes.
- Sum: the sum of all node values.

Input / Output Format

Input: One line: the tree in level order (values and N).

Output: Three lines: Inorder, Nodes, Sum.

Examples

Example 1

Input: tree = 4 2 6 1 3 5 7

Output:

```
Inorder: 1 2 3 4 5 6 7
Nodes: 7
Sum: 28
```

Explanation: In-order of this balanced tree yields the sorted values; there are 7 nodes summing to 28.

Example 2

Input: tree = 1 N 2

Output:

```
Inorder: 1 2
Nodes: 2
```

```
Sum: 3
```

Explanation: Root 1 with only a right child 2.

Constraints

- Values are integers; N marks a missing child.
- Build the tree level by level (left child then right child).
- An empty tree has no nodes and a sum of 0.

Sample Run

Sample Input

```
4 2 6 1 3 5 7
```

Sample Output

```
Inorder: 1 2 3 4 5 6 7
Nodes: 7
Sum: 28
```